PROJECT 3

STUDENT PERFORMANCE

Statistic for Business

PRESENTED BY MUNAWAR

STUDY CASE

Background

A high school wants to evaluate the effectiveness of a new learning method it will implement. The principal sets a target that the average student score must be above the competency standard of 8. If the population average is proven to be greater than this figure, then the learning method is considered successful. The school takes a random sample of 200 students from the population.

8.0

Target CGPA

Competency Standar

200

Sample Size

Student Sampled

α = 0.01

Significance Level

STEP 1 - DEFINE HYPOTHESIS

The null hypothesis states that the new learning method has no significant impact on student performance. We need to gather sample data to prove otherwise.

H_0 (Null Hypothesis)

$$\mu \leq 8$$

The new learning method has NO significant impact — the population mean CGPA does not exceed 8.0.

H_1 (Alternative Hypothesis)

$$\mu > 8$$

The new learning method IS effective — the population mean CGPA is greater than 8.0.

Why Used One-Sample t-Test?

- We use the sample mean to estimate the population mean (μ)
- Despite having n = 200 (large sample), the population standard deviation (σ) is unknown — so we cannot use a z-test
- The one-tailed (right-tail) test focuses the entire α on the right side, testing if μ > 8.0

STEP 2 - EXPLORATORY DATA ANALYSIS

Sample Mean

$\bar{x}$ = 7.63

Below target of 8.0

Std Deviation

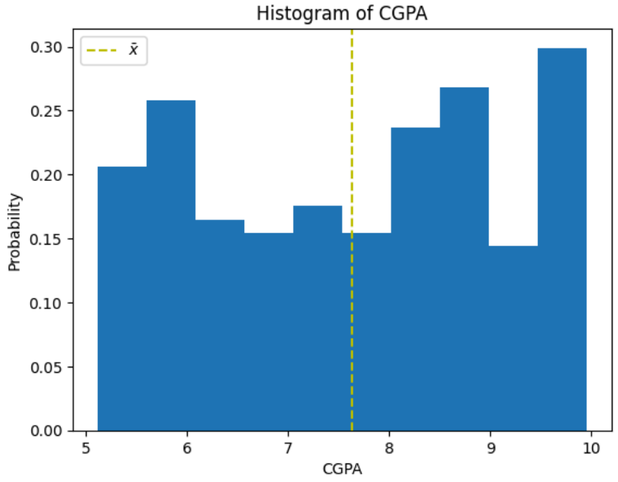
s = 1.49

Sample Standard Deviation

Difference

Δ = -0.37

Below competency standard



- The histogram displays CGPA per individual, not the distribution of the sample mean.
- We are interested in whether the population mean (μ) exceeds 8.0, not individual values.
- To formally test this, we need to measure how far ($\bar{x}$) is μ = 8 relative to the sampling variability — this is what the t-statistic captures.

STEP 3 - COMPUTE T-STATISTIC & CRITICAL VALUE

The t-statistic (t-statistic) is a number that shows how far the difference is between your sample data and the initial hypothesis, when measured using standard deviation.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad \Rightarrow \quad t = \frac{7.63 - 8}{1.49/\sqrt{200}} \approx -3.49$$

Where:

- $\bar{x}$ = sample mean
- μ_0 = hypothesized mean
- s = sample std dev
- n = sample size

t-Statistic

-3.4976

Critical Value

t_crit = 2.3452

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t_crit = 2.3452

STEP 4 - DECISION RULES & CRITICAL REGION

Method 1: p-value Approach

Reject H_0 if p-value < α

- p-value = 0.9997
- α = 0.01

0.9997 > 0.01 → Fail to Reject H_0

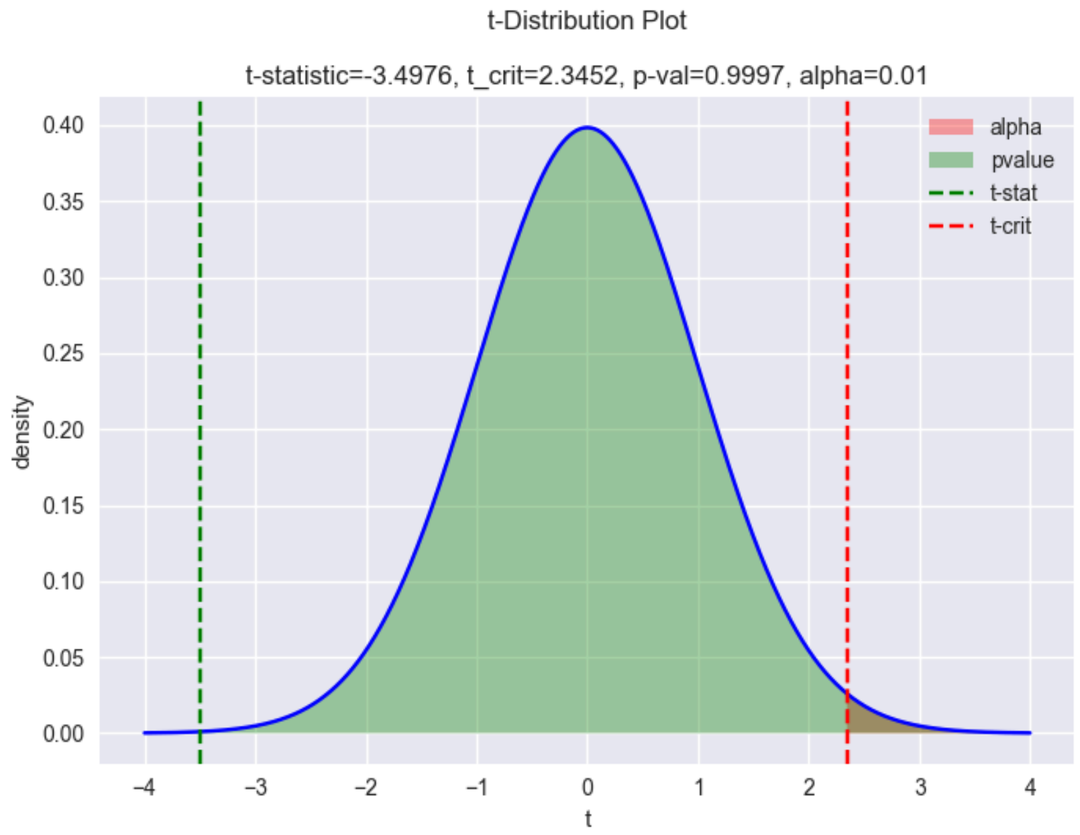
Method 2:Critical Value Approach

Reject H_0 if t-stat > t_crit

- t-statistic = −3.4976
- t_crit = 2.3452

−3.50 < 2.35 → Fail to Reject H_0

STEP 4 - DECISION RULES & CRITICAL REGION



- The plot shows the t-distribution under H_0. The **red region** (right of t_crit) is the rejection zone at α = 0.01.
- The **green region** represents the p-value - the probability of observing a t-statistic as extreme as ours, assuming H_0 is true.
- If the green region (p-value) is smaller than the red region (alpha), we reject H_0

STEP 5 - CONFIDENCE INTERVAL

95% Confidence Interval for Population Mean CGPA

$$7.42 \leq \mu \leq 7.84$$

But what happen with 8 that lies inside this interval?

- Since the p-value > alpha, we fail to reject H_0. This is consistent with our hypothesis test result — we fail to reject H_0.
- However, note that the CI approach uses a two-sided interval, while our hypothesis is one-tailed. This makes the CI less precise for this specific question. A one-tailed test is more appropriate when the direction of the effect matters, as it concentrates the entire α on one side and yields a sharper decision boundary.

CONCLUSION

- Based on the one-sample t-test at $\alpha = 0.01$ we fail to reject H_0.
- There is insufficient statistical evidence to conclude that the population mean CGPA exceeds 8.0.

RECOMMENDATIONS

1

Revise The Target
The sample mean CGPA of 7.62 is 0.38 points below the target of 8.0. The school should evaluate whether the 8.0 benchmark is realistic for this student population, or whether it needs to be recalibrated before the new learning method is rolled out more broadly.

2

Investigate Root Causes
The school may want to explore other variables (e.g., academic pressure, sleep duration, financial stress) that could be contributing to lower CGPA outcomes, as these factors may be affecting student performance independently of the learning method.

3

Investigate Root Causes
Run a pilot program with larger or more targeted samples for stronger evidence of the learning method's effectiveness over time.

THANK YOU FOR YOUR TIME

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DATA ANALYST PRESENTATION